

Evaluation Report

Hospital Readmission Risk Prediction

Dataset: cleaned_hospital_data.csv

Records: 50,000

Features: 30 raw → 46 final modeling features

Target: readmitted (0 = Not Readmitted, 1 = Readmitted)

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1. Class Imbalance Analysis

- Readmitted: 46,782 (93.6%) | Not Readmitted: 3,218 (6.4%) → 14.6:1 ratio.
- A naive classifier that always predicts readmission scores 93.6% accuracy without learning anything.
- Raw accuracy is a misleading metric — F1, ROC-AUC, PR-AUC, and Confusion Matrix used as primary metrics.
- Handling strategy:
 - Logistic Regression: `class_weight='balanced'` — auto-adjusts loss weights for the minority class.
 - XGBoost: `scale_pos_weight = not_readmitted_count / readmitted_count = 2,574 / 37,426 ≈ 0.069` — equivalent to class weighting.
- SMOTE tested as an alternative to `scale_pos_weight` (see Section 3).

2. Model Comparison

Four models evaluated on identical 80/20 stratified train-test splits:

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
Logistic Regression	0.882	0.993	0.879	0.933	0.962	0.997
Random Forest	0.993	1.000	0.993	0.996	0.997	1.000
XGBoost (Selected)	0.993	1.000	0.993	0.996	0.999	1.000
XGBoost + SMOTE	0.993	0.999	0.994	0.996	0.999	1.000

- Logistic Regression recall drops to 87.9% — the readmission boundary is non-linear; a linear model cannot fully capture it.
- Random Forest and XGBoost are near-identical — XGBoost edges ahead on ROC-AUC (0.999 vs 0.997).
- The 11.1-point accuracy gap between Logistic Regression and tree models understates the real clinical impact — the confusion matrix breakdown clarifies the cost in missed patients.

3. Imbalance Handling — scale_pos_weight vs SMOTE

Both methods applied to XGBoost only; the test set was always the original unmodified split:

Method	Accuracy	Precision	Recall	F1	FP	FN
XGB + scale_pos_weight	0.9930	1.000	0.993	0.996	0	~63
XGB + SMOTE	0.9930	0.999	0.994	0.996	~7	~56

- Decision: scale_pos_weight selected.
- SMOTE marginally improved recall (0.994 vs 0.993) but introduced ~7 false positives versus 0 for scale_pos_weight.
- scale_pos_weight requires no data augmentation; SMOTE synthesised 34,852 artificial samples, introducing noise risk without meaningful metric gain.
- In a clinical context, avoiding false positives (genuine patients incorrectly flagged as high-risk) is important — scale_pos_weight achieves this better.

4. Confusion Matrix Analysis

Results on 10,000 test samples (644 not readmitted, 9,356 readmitted):

Model	True Positive	False Positive	False Negative	True Negative
Logistic Regression	~8,179	~67	~1,177	~577
Random Forest	~9,293	~0	~63	~644
XGBoost	~9,293	~0	~63	~644
XGBoost + SMOTE	~9,300	~7	~56	~637

- Logistic Regression: ~1,177 false negatives — 1,177 high-risk patients discharged without intervention.
- Logistic Regression: ~67 false positives — 67 non-readmission patients unnecessarily flagged.
- Random Forest: 0 false positives and ~63 false negatives — a substantial improvement over the linear baseline.
- XGBoost: matches Random Forest on false negatives (~63) with 0 false positives — best overall clinical profile.

5. Clinical Cost Analysis

Assumptions: FN cost = ₹8,000 (missed readmission — unplanned re-hospitalisation) | FP cost = ₹1,500 (unnecessary care coordination for a stable patient).

Model	FP	FN	Total Cost (₹)
Logistic Regression	~67	~1,177	~9,517,500
Random Forest	~0	~63	~504,000
XGBoost	~0	~63	~504,000
XGBoost + SMOTE	~7	~56	~458,500

- Logistic Regression's cost is 94.9% driven by false negatives — 1,177 missed readmissions × ₹8,000.

- Both XGBoost and Random Forest reduce clinical cost by ~95% compared to Logistic Regression on the same 10,000-row test set.
- XGBoost is preferred over the SMOTE variant because it achieves the same FN count (63) with zero false positives.

6. ROC and Precision-Recall Curve Observations

ROC Curve:

- XGBoost AUC = 0.999 | Random Forest AUC = 0.997 | Logistic Regression AUC = 0.962.
- XGBoost and Random Forest both approach the top-left corner — near-perfect TPR across every FPR threshold.
- Logistic Regression's curve deviates visibly at high sensitivity — full recall requires accepting a higher false positive rate.

Precision-Recall Curve:

- XGBoost AP = 1.000 | Random Forest AP = 1.000 | Logistic Regression AP = 0.997.
- All three models far exceed the 0.936 baseline (readmission prevalence).
- XGBoost and Random Forest maintain precision near 1.0 through maximum recall — threshold tuning is largely unnecessary.
- Logistic Regression's PR curve drops past ~90% recall, confirming the linear model struggles at the boundary of the minority class.

7. SHAP Feature Importance (XGBoost)

Rank	Feature	Direction of Effect
1	previous_emergency_visits	Higher count → Higher readmission risk
2	previous_admissions	Higher count → Higher readmission risk
3	total_risk_history	Higher score → Higher readmission risk
4	followup_compliance_score	Lower score → Higher readmission risk
5	blood_pressure	High category → Higher risk
6	has_secondary_diagnosis	Present → Higher risk
7	hospital_stay_days	Longer stay → Higher risk
8+	patient_age, bmi, glucose_level, heart_rate, hospital_bill_amount, insurance_type	Minimal contribution

- Prior healthcare utilisation dominates. The three strongest predictors — emergency visits, previous admissions, and combined risk history — all describe a patient's historical interaction with the system.
- Compliance matters. followup_compliance_score ranked fourth; patients who do not follow post-discharge appointments are at meaningfully higher readmission risk — and this is directly actionable through care coordination.
- Clinical vitals contributed minimally. patient_age, bmi, glucose_level, and heart_rate showed near-zero SHAP values. This is consistent with the synthetic nature of the dataset, where vitals were not embedded as strong predictors.

- Secondary diagnosis and blood pressure had moderate effects above the noise floor, consistent with clinical reasoning.

8. Final Model Selection — XGBoost

- Highest ROC-AUC (0.999) among all four models — best probabilistic separation between readmitted and non-readmitted patients.
- Perfect precision (1.000) — zero non-readmission patients incorrectly flagged as high-risk on the test set.
- Near-perfect recall (0.993) — only ~63 readmitted patients in 10,000 were missed by the model.
- `scale_pos_weight` handles class imbalance without synthetic data generation, keeping training data authentic.
- Gradient boosting captures non-linear feature interactions (e.g., `previous_emergency_visits` × `followup_compliance_score`) that Logistic Regression cannot learn.
- Natively supports SHAP explainability via TreeExplainer — critical for clinical staff trust in model predictions.

9. Risk Stratification

Post-prediction, the readmission probability is translated into actionable clinical tiers:

Probability Range	Risk Tier	Recommended Clinical Action
< 0.40	Low Risk	Routine 30-day follow-up. Standard discharge instructions.
0.40 – 0.70	Medium Risk	Follow-up appointment within 14 days. Review discharge plan.
≥ 0.70	High Risk	Early review within 48–72 hours. Assign care coordinator. Confirm medication adherence.

- Test set distribution: High Risk — 9,291 (92.9%) | Medium Risk — 0 (0.0%) | Low Risk — 709 (7.1%).
- The binary split between High and Low reflects the model's very high confidence on this synthetic dataset — probabilities cluster near 0 or 1 with no middle-ground uncertainty.
- In real clinical deployment, a wider distribution across tiers is expected as the model encounters more ambiguous cases.